

Particle Methods

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Final Project

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A stochastic cellular automata model of chemotherapy effects on tumour growth

Project overview

For my final project in this particle methods course, I decided to focus my attention on a topic related to the biological field and in particular on what are the effects of chemotherapy on tumor growth, clearly giving priority to aspects related to the modulation and simulation of this phenomenon, more than to the actual medical-biological details. In fact, the idea behind this project is to demonstrate how a simple model such as the one I am going to describe is actually very useful and for this reason much used to describe complex phenomena as the one mentioned above.

In these terms, through this project, I want to introduce the SCAM model, namely a stochastic cellular automata model. It consists of a “classical” cellular automata model, where the stochasticity is represented by the probabilistic rules that manage the interactions between cells in the tissue under consideration. In order to do this, the idea was to divide the work into two parts, the first in which I describe the growth of the tumor in all its parts, thus looking at the different layers of particles that constitute it and the interactions between tumor cells and immune cells. In the second part, on the other hand, I focus on what is the real object of the project, which is the simulation of the effects of therapy on the tumor, thus going to analyze how the different parameters related to the chemotherapy act on the formation of the tumor mass.

Before introducing you to these two parts of the project, however, let me describe the implementation steps of the model, for which I used the programming language Matlab, whose code can be find in the matlab file '*SCAM_model*'.

Implementation of the model

The model is structured as a two dimensional lattice with periodic boundary conditions in which each side consists of a number of cells equal to ' n_cells ' and where a finite set of possible states is provided for each cell. Specifically, in this case, seven possible states for a cell are assumed:

- State 0 = normal cell or empty space (NoC or ES)
- State 1 = proliferative tumor cell (PC)
- State 2 = quiescent cell (QC)
- State 3 = necrotic cell (NeC)
- State 4 = unstable state (US)
- State 5 = immune cells (IC)
- State 6 = dead cell as a result of immune system or therapy (DC)

Given the high amount of information that the model requires to store for each cell, the original idea was to create a structure that provided different fields for each of them with the various reference values, but this implementation would have entailed very high computational costs, since we have around forty thousand cells in the domain ($L = n_cells = 200$). Consequently I adopted a different method, namely that of creating a grid for each piece of information to be memorized so as to make the model and the simulation itself more efficient.

Firstly I defined the function '*initialize_grid*' that is called at the beginning of the code to initialize the different grids and to define the initial configuration of the cells in the lattice. We therefore have the '*cells*' grid which stores all the cell states with values ranging from 0 to 6 as said before. Subsequently we have the two grids '*drug_resistant_cells*' and '*n_dead_steps*' which respectively contain the resistance state of the cells and the number of steps after which each cell dies due to the therapy. In particular, the first one can be defined as a boolean type grid since it assigns only two values at each cell regardless of cell state, thus value 0 if the cell is drug sensitive (DSC) or value 1 if it is drug resistant (DRS), the second one instead assigns a random value at each cell based on the treatment variables defined in the parameters section (this is the only grid that is not updated at each iteration unlike the others). In the end we have the last two, hence the '*age_cells*' and the '*survival_steps*' grids, which are defined as 'counter grids' since they update the age of cells and the number of steps in which the cell does not die respectively, starting clearly by value zero.

Once the initial elements of the model had been defined, as well as the parameters used in the simulation, which are divided into growth variables (parameters used for the first part of the project) and therapy variables (parameters used to simulate the effects of chemotherapy), I proceeded with the definition of all the functions used in the construction of the model, starting by those relating to the definition of the neighborhood of each cell. The paper in this sense assumes the use of the Moore neighborhood for the simulation, but since at each iteration I have to compute the radius of the tumor mass given by the distance of the external edge of the tumor from the center, I decided to refer for this particular calculation to the Von Neumann neighborhood instead. In fact, in order to compute this quantity, I have to identify at each

time step the set of cells which composed the boarder of the tumor mass. Therefore, in order to avoid redundancies in the calculations, I need to identify those cells that have at least one empty space in their Von Neumann neighborhood, which unlike Moore's does not look at the 8 adjacent cells but thinks about the only four cells that touch the edges of the considered cell, as we can see in the following figures:

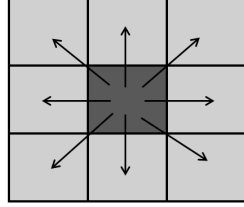


Figure 1: Moore neighborhood

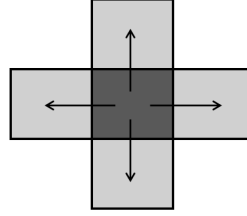


Figure 2: Von Neumann neighborhood

Immediately after these two functions, we have instead a series of other functions that describe the quantities and probabilistic rules that come into play in the main function that defines the interactions between different cells. I am not going to list and present all the individual formulas used here since there are quite a few but they are still formulas used to calculate for example the probability of division for proliferation, the thickness of the different tumor layers, the average radius of tumor, as well as the growth rate of recruiting for immune cells and the exponential treatment response curve for the chemotherapy.

As mentioned, all the elements just described are called within the main function '*update_grid*', which as anticipated describes the cell transition rules, that is, all those rules that define the interactions between different cells and describe their behaviors within the domain. First it should be said that I decided to implement a random update of the cells in the lattice so as to avoid the creation of any artifacts for this reason and to ensure that the simulation is as random as possible. Immediately after that, I defined the behaviors and possible interactions for each cell, dividing right into the individual cases through an if statement with six different cases, one for each cell type in the tissue. So, I summarize below the main behaviors of some cells within the system:

- QC cells = each quiescent cell changes its state base on the distance from the center. In particular, it turns into a NC due to lack of nutrition or turns into a PC due to accessing sufficient nutrient, going to form the various layers of the tumor mass.
- DC cells = dead cells are the result of therapy and they merely change state when they have more than five PCs in their neighborhood, turning into normal cells.
- PC cells = each proliferative tumor cell can proliferate with a probability of division, which is a function of time and the distance of dividing cell from the centre of the lattice. At each time step, every PC is checked to see if it can be divided. If there is at least one NoC or ES in its neighborhood the first daughter cell will remain in the same position of its parents and the second one will be randomly put in one of the ESs of its neighbourhood. In this case, however, it is also necessary to observe whether the cell under consideration is drug resistant or not because in that sense there are two different forms of mitosis. In fact replication of DSCs is always supposed to be symmetric, i.e. a proliferation division results in the creation of two daughter cells of the same type, while DRCs will

be assumed to sustain either symmetric or asymmetric (creation of two daughter cells of different types), i.e. one DRC daughter and one DSC daughter will be born from the replication.

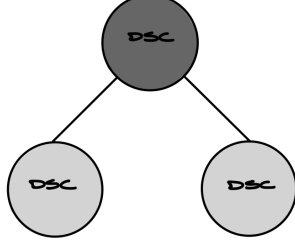


Figure 3: Example of symmetric replication

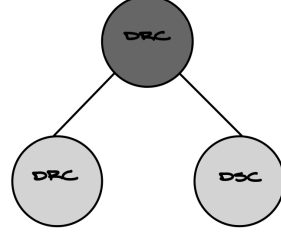


Figure 4: Example of asymmetric replication

- IC cells = the immune cells move toward the center of the tumor because we have assumed that debris from dead tumor cells may help the immune system recognize its target. In order to do that, I implement a walk for which at each time step every IC is checked to see if the closest cell to the center of its neighborhood is an empty space. If this is verified, the cell will take that place with a certain probability, otherwise it will move randomly in one empty space of its neighborhood. These cells continue to walk until they don't find a PC cell. When this happens, they try to "eat" all the PC cells in its neighborhood with a certain probability and if they fail, they simply move towards an empty space. In addition, in this model immune cells are also subject to a recruitment process. Since an increase in the number of tumour cells will stimulate the immune system, the positive nonlinear growth rate of recruiting ICs defines a system for which at each time step a certain number of new immune cells are recruited at the corners of the lattice based on number of successes and failures in killing proliferative cells (see red circles in the picture). As we can observe in the following figure, although in this case I manually increased the probability of success ' r_i ' in order to have a clearer demonstration of the process, when the difference between successes and failures is positive the recruitment process is activated:

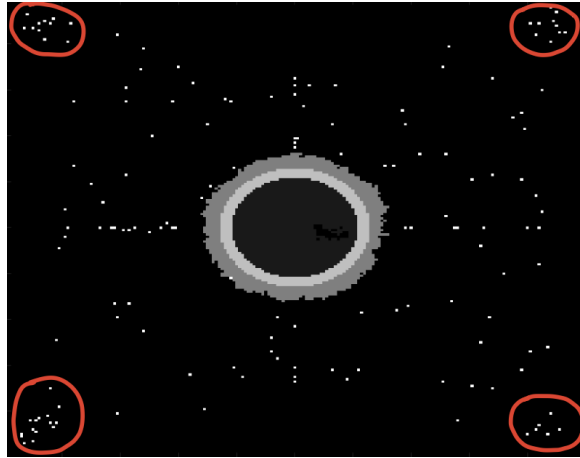


Figure 5: Recruitment process of immune cells

Other cells, on the other hand, just sit still and increase their age over time, which clearly happens for all the cells listed above as well. When the therapy is applied, however, new dynamics come into play and these are common for all types of cell (except necrotic cells and DRCs). In particular, based on the values of parameters initially defined, any cell can be killed by the chemotherapy with a certain probability and if this doesn't happen, a new survival step is added to the cell through the relative grid. Clearly, in this simulation, those cells that are not drug resistant sooner or later die because of the therapy, and therefore when the survival steps exceed the relative threshold defined by the grid ' n_dead_steps ', these cells turn into unstable state first and dead cell later.

To conclude, however, we clearly find the main loop for the simulation. Given the following time structure for the therapy

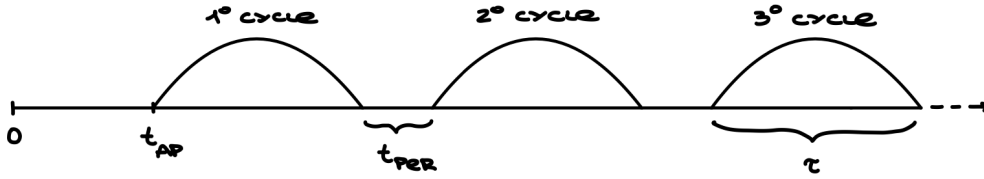


Figure 6: Example of time structure of chemotherapy with 3 cycles (where t_{ap} = start time of therapy, t_{per} = rest period between one cycle and the next, τ = cycle duration and n_d = number of cycles)

it goes to apply the dynamics of chemotherapy only in those steps included in the therapy cycles, instead allowing the dynamics linked to tumor growth to develop normally in all the others. In order to do that, I set the parameter ' $apply_therapy$ ' of the ' $update_grid$ ' function as 'false' when the therapy does not have to be applied, while as 'true' when it should have its effect on the tumor. At each time step, however, it checks whether it should apply the immune cell recruitment process I was talking about before and also updates the radius of the tumor mass and other relative quantities which are then used in the main function as we have seen. Clearly as a consequence it also updates the plot showing the dynamics of the cells in the tissue.

First part: tumor growth

In the first part of this project I focused on simulating the growth of the tumor mass as mentioned, going to observe the process of formation of the different layers of tumor cells. As suggested in the paper, the initial configuration of the domain consists of just one proliferative cell in the center of the lattice and a small number of immune cells ($0.001 \times n_cells^2$) randomly located in the domain itself. The general idea of this first part is to observe how the tumor is created over time and after some iterations you can already see the different layers of cells forming until it reaches a complete shape like the one shown in the figure below. In particular we can understand that the tumor has reached maximum size when its radius equals ' R_{max} ', which represents the maximum tumor extent and it is set equal to 37.5. Clearly, by setting a lower value for this parameter leads to having a smaller tumor mass and consequently a tumor that will be composed by a lower number of tumor cells.

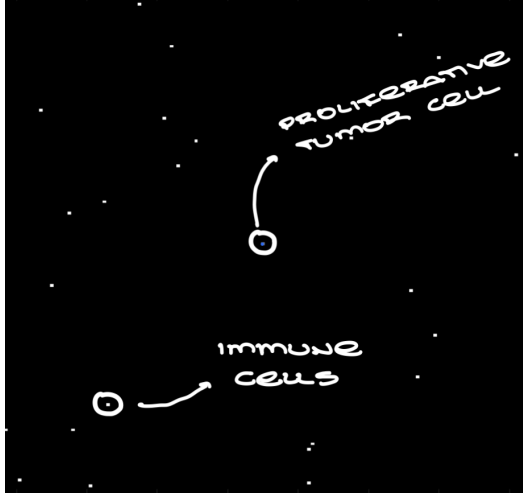


Figure 7: Initial configuration

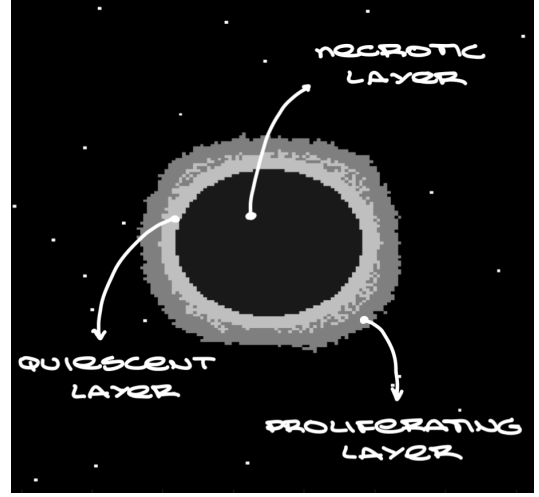


Figure 8: Final configuration

Looking at the figure 8 on the right, we can clearly see the three different layers that characterized the tumor. On the outside we find the medium grey layer formed by the proliferative cells, the one composed by cells that reproduces themselves at each time step since they are closer to the nutrient represented in our model by cells in state 0. In the middle we can find instead the quiescent layer composed by QC cells, which represents those PC cells which are not close to the nutrient but also not too close to the center of the tumor mass, so basically they are in an intermediate state. At the end we have the necrotic layer represented by the dark grey mass. Necrotic cells accumulate in the inner part of the tumour over time go to form a mass of dead cells which cannot be affected by chemotherapy. In fact, the state of necrotic cells cannot be influenced by tissue's repair process, and it is assumed that surgery is the only way that necrotic cells can be removed from the studied tissue.

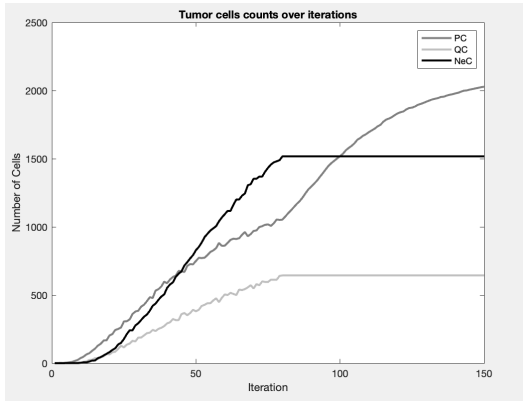


Figure 9: Total number of tumor cells with $R_{max} = 37.5$

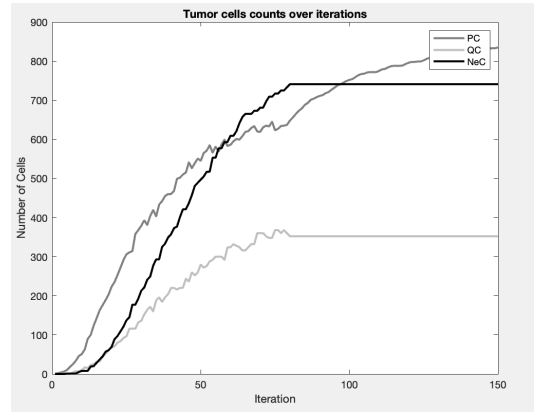


Figure 10: Total number of tumor cells with $R_{max} = 25$

If we plot instead the evolution of the number of tumor cells, thus proliferative, quiescent and necrotic cells, we can observe that the smaller the value of ' R_{max} ' the lower the number of cells. Contrary to what I thought, maximum tumor extension though does not occur in fewer iterations. These in fact remain almost identical and what actually changes is only the number

of cells and this is probably due to the fact that there will also be fewer cells proliferating, thus implying a slowness in the process of mitosis. However, also in this figures it is easy to identify when the mass reaches its maximum extension, thus when the number of necrotic cells remains at a certain level without continuing its upward trend and how we can see it seems to be around 75 iterations in both cases. Similar results are also obtained by reducing the probability of division for PC cells, but we see this better later in the second part when we discuss the effects of chemotherapy.

What happens instead if we change the values of ' a ' and ' b ', that is, the base necrotic thickness and the base proliferative thickness (living tumor), set equal to 0.42 and 0.53 respectively ?

By starting from the value of the parameter ' a ' we can observe the following graphs:

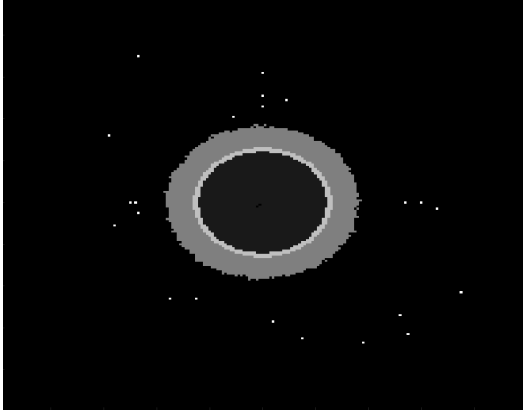


Figure 11: Final configuration of tumor mass with $a = 0.2$

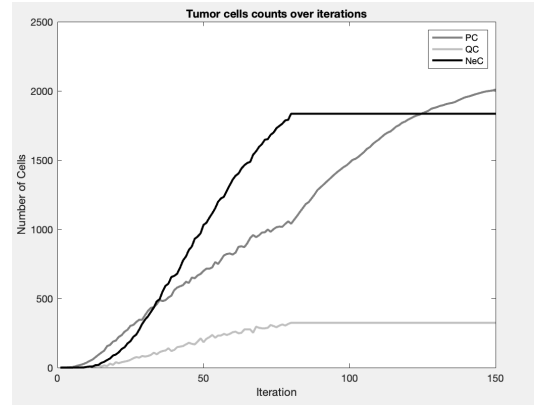


Figure 12: Total number of tumor cells with $a = 0.2$

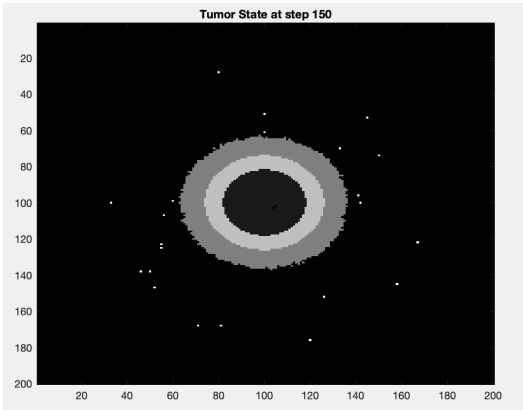


Figure 13: Final configuration of tumor mass with $a = 0.8$

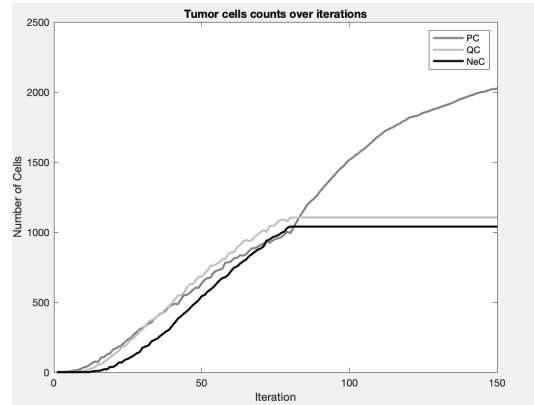


Figure 14: Total number of tumor cells with $a = 0.8$

Although in theory we are operating on the necrotic layer, it is evident how the tumor mass presents a very thin quiescent layer if we go to consider a low value of the parameter, presenting instead a much thicker layer if we consider a higher value, and this finding can be observed also

in the total number of quiescent cells at the end of the simulation. Clearly, leaving the value of the maximum radius ' R_{max} ' fixed, the other two layers as a consequence will also undergo some modification, in particular the necrotic one which turns out to be particularly small in the case where we set a high value for the parameter ' a '.

Different results on the other hand if we change the value of parameter ' b '. In fact, looking at the following plot, we can notice that for small values of this parameter, the tumor mass is not able to reach its maximum extension, probably due to the fact that the mitosis process is negatively affected since we have less PC cells. On the opposite side instead, by increasing its value we observe a thicker proliferative layer and a larger number of PC cells than the base case, reaching nearly one thousand more cells and leading to a more efficient proliferation process than seen with a parameter $b = 0.2$

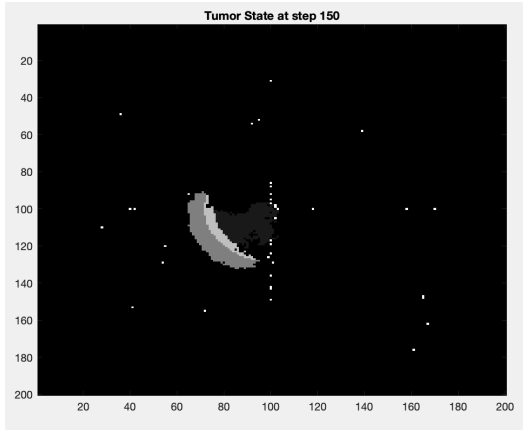


Figure 15: Final configuration of tumor mass with $b = 0.2$

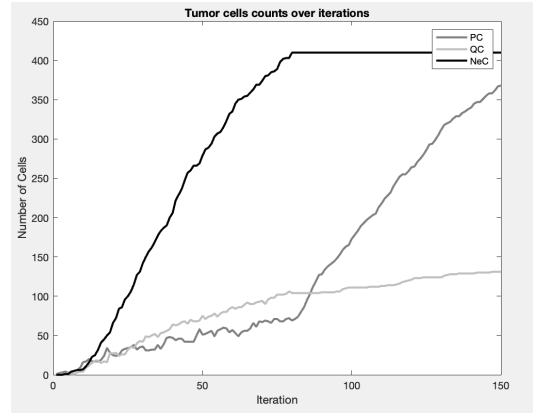


Figure 16: Total number of tumor cells with $b = 0.2$

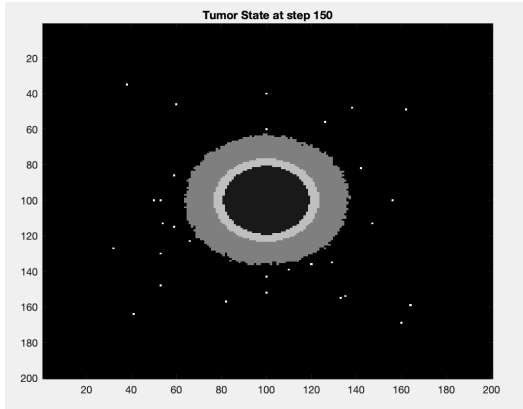


Figure 17: Final configuration of tumor mass with $b = 0.8$

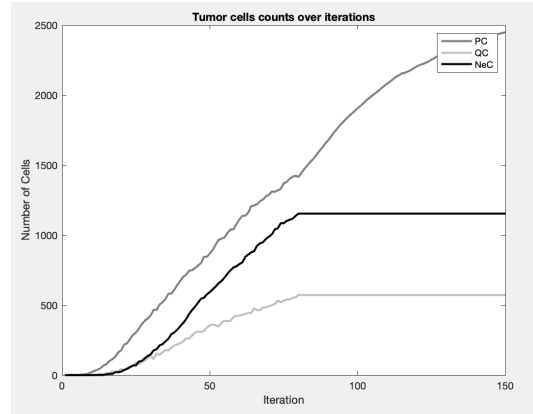


Figure 18: Total number of tumor cells with $b = 0.8$

Second part: chemotherapy effects

In this second part of the project instead I decided to go into more detail about what is the subject of the paper, which is the analysis of the effects of chemotherapy on tumor growth. In this sense, for the sake of clarity, it must first be said that we assume that chemotherapy can affect tumour growth dynamics via three mechanisms:

- decreasing the probability of proliferation of PCs
- reducing tumour carrying capacity
- inducing cell death into tumor cells, through modification of therapy variables

Depend on the chemotherapy drug or type of administration chosen for a patient, one or all of the mechanisms may be activated. Moreover, the combined therapy may be considered by activating multi-mechanisms simultaneously. We will now analyze all three mechanisms and a few variations of them in order to understand which is the most efficient one.

NB: just to clarify, in the following plots the white dots distributed in the lattice represent not only the immune cells (which, however, given the low probability of reproduction and the very small number in the initial configuration tend to disappear after a few iterations) but also the cells that died due to the therapy.

1) Chemotherapy simulation via regulations of the base probability of division of PCs

The probability of proliferation ' ρ_0 ' for PC cells represents the probability at which proliferative cells reproduce themselves through mitosis. It is the base probability which goes to influence the probability of division ' b_r ' so defined:

$$b_r = \rho_0 * \left(1 - \frac{r}{R_{max} - K_c}\right)$$

In particular, although the value of ' ρ_0 ' is initially fixed at 0.7, in order to consider the effect of therapy in the simulation, during the different cycles the base probability of proliferation is modified in this way:

$$\rho_0 = \frac{\rho'_0 * \gamma'_{PC}}{n_d^{1/n_{dead}}}$$

where ρ'_0 is the base probability of division in the absence of chemotherapy and γ'_{PC} shows the drug-induced resistance of PC cells. Therefore, according to this formulas, reducing ' ρ_0 ' will significantly decrease the tumour volume, since this action will affect negatively the probability of division, thus the process of mitosis making it less efficient.

As we can see in the plots below, the volume of the tumor mass is smaller if we consider lower values of this parameter and the difference is clearly visible even decreasing the probability value by a small amount compared with the base case (for this simulation we are considering the following therapy variables: $t_{ap} = 80$, $t_{per} = 5$, $\tau = 15$ and $n_d = 4$). Especially smaller will also be the necrotic layer of the tumor and consequently this will allow for less incisive surgery

in reality since a smaller portion of the mass will have to be removed. Obviously this will also affect the total number of tumor cells, which will be lower than in the case with $\rho_0 = 0.7$.

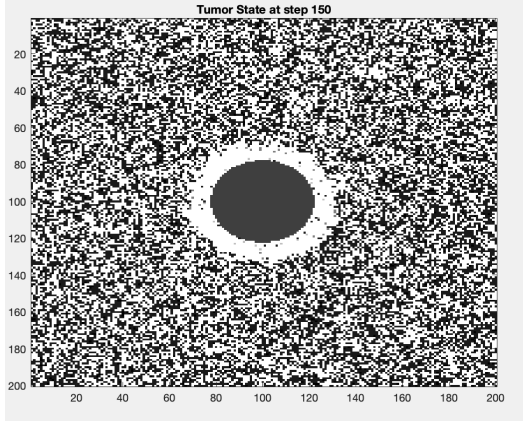


Figure 19: Final configuration of tumor mass with $\rho_0 = 0.7$

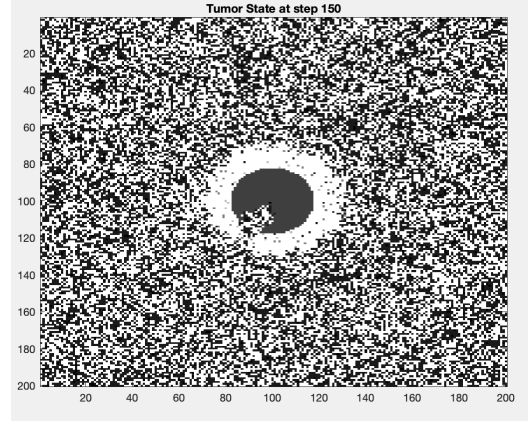


Figure 20: Final configuration of tumor mass with $\rho_0 = 0.5$

2) Chemotherapy simulation via carrying capacity regulations

The concept of carrying capacity is typically used in ecology to describe the maximum population size of a species that an environment can sustain indefinitely, given the availability of resources, space, and other environmental constraints. When applied to tumor growth, the concept of carrying capacity can help explain how a tumor grows and stabilizes within a host organism. In this sense, the change in the ' K_c ' parameter consequently involves a change also in the structure and especially in the size of the tumor mass, considering that its value can be set equal to a number between 1 and half of the maximum radius ' R_{max} ', thus 18 in the base case.

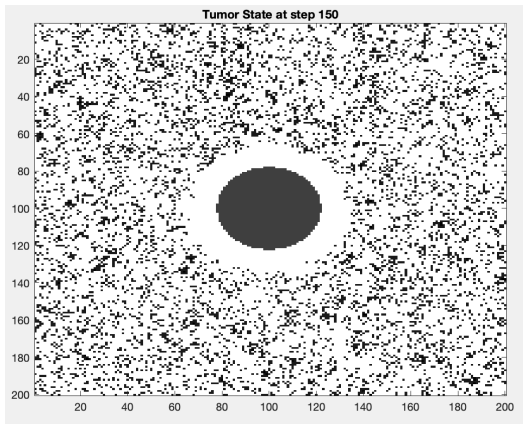


Figure 21: Final configuration of tumor mass with $K_c = 1$

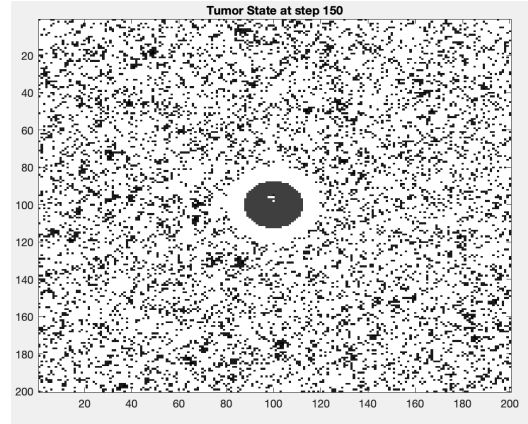


Figure 22: Final configuration of tumor mass with $K_c = 18$

Looking at the figures, they show the differences we were talking about (maintaining the same parameters used in the previous case with a base probability of proliferation equal to $\rho_0 = 0.7$).

In particular, an increase in ' K_c ' (i.e. reducing the carrying size of the tumor) leads to the survival of more normal cells in the host tissue and consequently in the development of fewer cancer cells. In fact, looking again at the formulas for the probability of division for PC cells, we can see that the parameter, that is placed in the denominator, negatively affects the probability itself with an increase in its value, result in a reduction of the tumor mass. This is why a treatment that can lead to ' $K_c = 18$ ' is suggested.

3) Effect of initiating therapy time

This kind of variation in the simulation is very intuitive and clearly changing the step at which chemotherapy starts leads to different results with reference especially to the growth of the tumor mass. With a value of t_{ap} very low, the therapy starts early and therefore it will result in the number of tumor cells starting to decrease earlier than in the case where the step of initiating chemotherapy is instead set later in time.

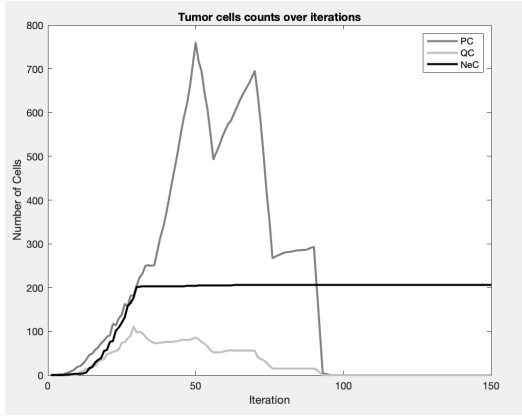


Figure 23: Total number of tumor cells with $t_{ap} = 30$

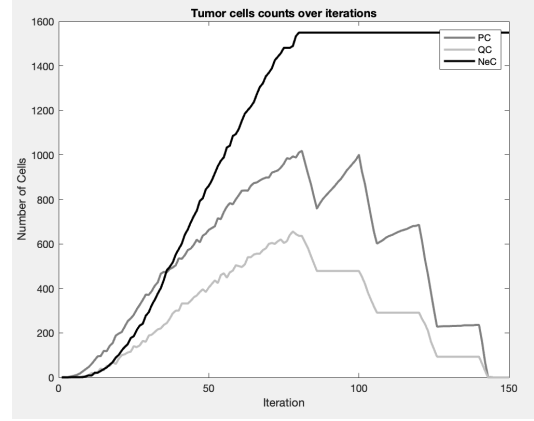


Figure 24: Total number of tumor cells with $t_{ap} = 80$

It is therefore easy to see how a therapy that starts very early already almost at the diagnosis of the tumor itself can lead to better results, avoiding the formation of a large tumor mass and reducing from the very beginning the risk of having too large a number of tumor cells to manage (in the graphs initiation of therapy is visible at the point where necrotic cells begin to be constant over time).

4) Effect of time interval between infusions

The time interval between infusions is represented by the variable ' t_{per} ' and indicates the time period of rest from therapy. In fact, during this intervals the parameter ' $apply_therapy$ ' is set to false in order to avoid the application of the treatment on the tumor and consequently this leads to a continuation in the formation of the tumor mass during this period.

As is visible from the plots below, the longer the time between infusions, i.e. between cycles, the longer PC cells can continue to proliferate. Longer increasing trends for proliferative cells can be noted in the second case with a duration of the rest period set equal to 20 and in the same time longer longer periods in which QC cells remain constant. Clearly, the greater the value of this parameter, the greater the number of iterations necessary to achieve complete "destruction" of the active tumor mass, therefore of the PC and QC cells in the tissue.

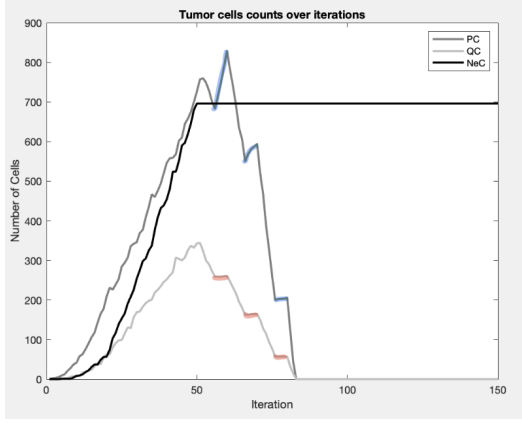


Figure 25: Total number of tumor cells with $t_{per} = 5$

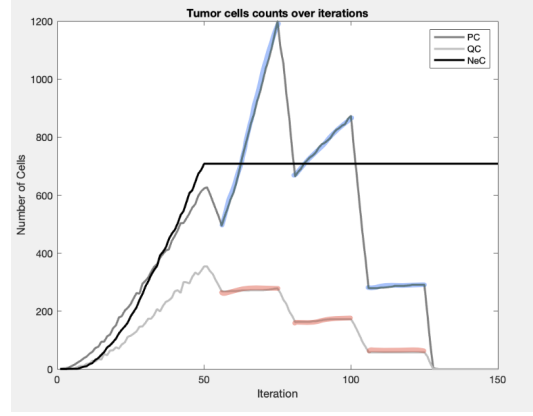


Figure 26: Total number of tumor cells with $t_{per} = 20$

5) Effect of duration of each cycle

Completely opposite case to the one just seen is the one in which we operate on the time length of each cycle, i.e. when we change the value of the parameter ' τ '. In this case the situation is exactly the opposite, having longer decreasing trends for PC cells when the value of the variable is high and shorter trends when the value is low instead. In fact, as can be seen from the graphs, the longer the time between the start of the cycle and the end of the same, the longer the therapy can have an effect on the tumor cells.

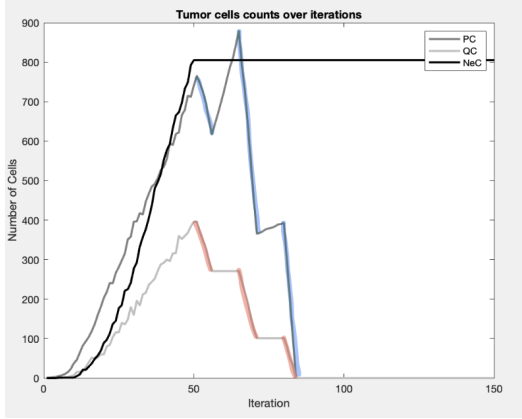


Figure 27: Total number of tumor cells with $\tau = 5$

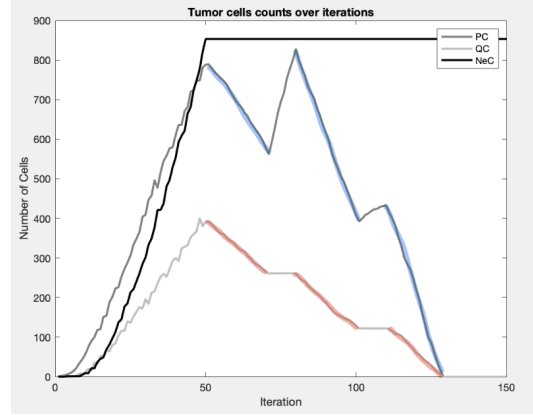


Figure 28: Total number of tumor cells with $\tau = 20$

Although it is rightly expected that with longer cycles of therapy the number of iterations necessary to defeat the tumor is lower, in the graphs this does not occur since the number of time steps before death due to the treatment for each cell is set in randomly based on the value of ' τ ' itself and consequently by increasing this parameter, the range for the values of the grid ' n_{dead_steps} ' also increases.

If we manually set the value of ' τ ' within this formula so as to always have the same range for these values, we can instead notice how the number of iterations necessary is reduced as the

```

% initialize n_dead values (number of time steps before death
% due to the treatment)
for i = 1:L
    for j = 1:L
        n_dead_steps(i, j) = randi([1, (tau*n_d)]);
    end
end

```

Figure 29: Formula for the initialization of the number of survival steps for each cell

value of the length of each cycle increases, as actually happens in reality, confirming what was said on a theoretical level.

6) Effect of number of cycles

The number of cycles has a lower influence than the parameters seen previously on the formation of the tumor mass, but despite this we can observe how a greater number of cycles involves a longer therapy process which leads to longer and less pronounced decreasing trends for the proliferative cells over time. However, even in this case this behavior is poorly justified on a theoretical level and is due to the probabilistic rules of the model, in particular again by the function that initializes the values of the dead steps, as it also depends on this parameter. In any case, the different cycles of chemotherapy are clearly visible from the graphs as we can see from the following examples where I have considered two and four treatment cycles respectively:

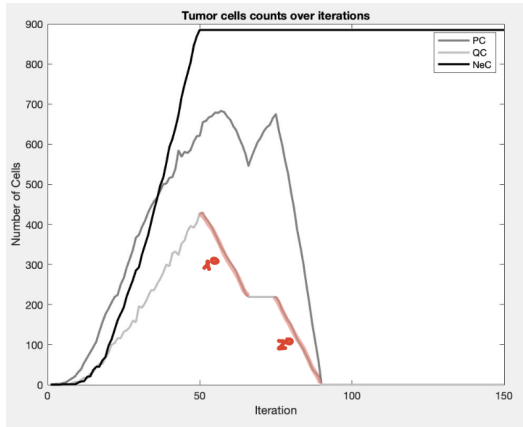


Figure 30: Total number of tumor cells with $n_d = 2$ cycles

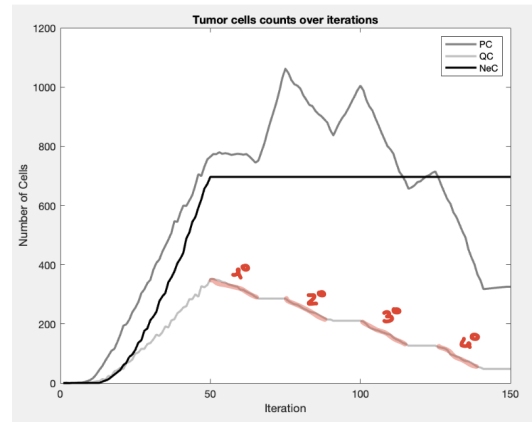


Figure 31: Total number of tumor cells with $n_d = 4$ cycles

Simulation results

To summarize what was seen during the various simulations carried out, the parameters linked to the therapy that influence the formation of the tumor mass are many and different. Those that affect the probability of division for PC cells allow different volumes of the tumor to be obtained and therefore a different number of tumor cells in the tissue, which is why they are the preferred parameters on which to act. At the same time, a certain influence is also given by those parameters linked to the temporal structure of the chemotherapy itself, i.e. the number of cycles, the duration of each cycle and the rest period between one infusion and another, without clearly forgetting the start time of therapy which is probably the most influential of all these.

However, a separate mention must also be made of all the other therapy variables of the model which in our case are initially fixed. I'm talking about the coefficients of resistance to therapy of different cell types, as well as their death rates. These are values which, if increased or decreased, clearly modify the results of the simulations. In particular, by decreasing the resistance coefficients we will have a system in which the therapy has more effect on individual cells and therefore where fewer iterations will be necessary for tumor defection. Instead, with reference to death rates, if these are decreased then it will be more difficult for the chemotherapy to "eat" the tumor cells and consequently a longer period of therapy will be necessary to reduce the tumor mass.

Conclusions

The idea of this project, as initially said, was to show how an apparently simple and basic model like the one described turns out to be actually very useful from many points of view for the simulation of a complex phenomenon such as that of tumor growth and of the effects of chemotherapy on it. In particular, in this work I was very surprised by the fact that the cellular automata model allows us to consider even very small details of the different phenomena and it is for this reason that it is often preferred over other models. In fact, as stated in the paper, many drug designers rely on this model to understand which parameters most influence the growth of tumors and therefore to identify those elements that cannot be missing in the creation of a drug or therapy aimed precisely at defeating tumors.

Furthermore, an even more interesting thing that I discovered while working on this project is the fact that there is never a limit to the complexity of a model. There is always the possibility of adding details to the simulation and making it as precise as possible, thus getting closer and closer to what reality is. With reference to this model in particular, it is curious to see how hundreds of different probabilistic rules can be defined which therefore allow us to make fewer assumptions and instead follow more of the actual dynamics of the phenomenon in order to create a very accurate simulation. Therefore several improvements and further ideas can be implemented to this model that I have just described to you, this is just the beginning !

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